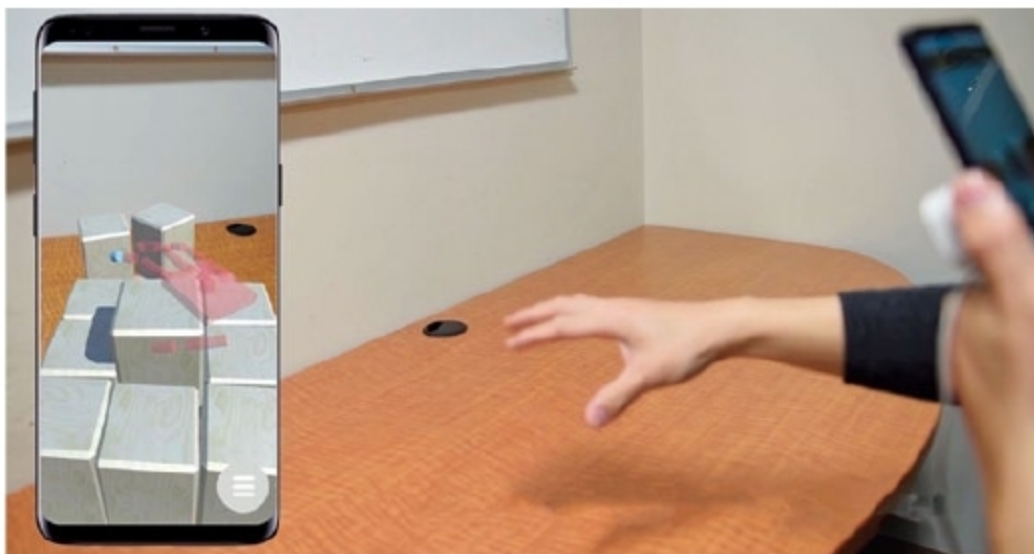


the idea for Portal-ble's hands-on interaction grew out of frustration with AR apps like Pokémon GO. AR apps use smartphones to place virtual objects (like Pokémon characters) into real-world scenes, but interacting with those objects requires users to swipe on the screen. And, swiping is just not a satisfying way of interacting.

The platform makes use of a small infrared (IR) sensor mounted on the back of a phone. The sensor tracks the position of people's hands in relation to virtual objects, enabling them to pick objects up, turn them, stack them or drop them. It also lets people use their hands to virtually paint onto real-world backdrops. As a demonstration, Huang and his students used the system to paint a virtual garden into a green space on Brown's College Hill campus.



Using a smartphone, people can use their hands to manipulate virtual objects on real backgrounds, thanks to an AR system developed at Brown University (Credit: Brown University)

## Engineers develop way to know liars' intent

Dartmouth engineering researchers have developed a new approach for detecting a speaker's intent to mislead. The method's framework could be applied toward evaluating fake news, among other uses.

Although previous studies have examined deception, this is one of the first to look at a speaker's intent. Researchers put forward an argument that while a true story can be manipulated into various deceiving forms, the intent, rather than content of the communication, determines whether the communication is deceptive or not. For example, the speaker could be misinformed or make a wrong assumption, meaning he or she made an unintentional error but did not attempt to deceive.

"Deceptive intent to mislead listeners on purpose poses a much larger threat than unintentional mistakes," said Eugene

Santos Jr., co-author and professor of engineering at Dartmouth. "To the best of our knowledge, our algorithm is the only method that detects deception and at the same time discriminates malicious acts from benign acts."

The researchers have developed a unique approach based on artificial intelligence (AI) and the resulting algorithm that can tell deception apart from all benign communications by retrieving universal features of deceptive reasoning. However, the framework is currently limited by the amount of data needed to measure a speaker's deviation from their past arguments; the study used data from a 2009 survey of hundred participants on their opinions on controversial topics, as well as a 2011 dataset of 800 real and 400 fictitious reviews of the same twenty hotels.

## Supercomputers can now track drug-resistant viruses



Supercomputers at Texas Advanced Computing Center (Credit: TACC)

As the days suddenly transition from summer warmth to a winter chill, more and more people get sick. The colder months signify that seasonal bugs are in full flux.

The flu ranges from an inconvenience for some to life-threatening for others. For elderly people, pregnant women and young children, the spread of the flu is ominous.

However, if we knew more about these viruses, researchers believe we could learn their weaknesses and stay ahead of them. Now, scientists are using supercomputers at Texas Advanced Computing Center (TACC) to find ways to stop viruses in their tracks.

Peter Kasson, associate professor of molecular physiology and